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Therapeutic Applications of *Nigella Sativa* and its Metal Nanoparticles

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Abstract: *Nigella sativa* is an annual flowering plant belongs to the family Ranunculaceae commonly known as Black seed, Black cumin or Kalonji. It is considered as one of the most miraculous healing medicines existed as a remedy for all diseases specified in varied traditional system of medicine and food. Phytochemically, it is enriched in fixed oil, terpenoids, essential oil, alkaloids etc and few key chemical constituents such as thymol, carvone, thymoquinone, nigellicimine, nigellicine, dithymoquinone and thymohydroquinone. Therapeutic properties of this plant are due to the presence of thymoquinone which is one of major active component and has different beneficial properties. Ample of phytochemical, pharmacological and clinical researches have been executed on this plant which may include antidiabetic, anticancer, immunomodulator, analgesic, antimicrobial, anti-inflammatory, bronchodilator, hepato-protective, renal protective, gastro-protective, and antioxidant properties etc. This plant also been studied for nanoparticle synthesis via green route with varied biological applications. Considering above all these aspects, this plant catches the attention of research analyst to tactic the utility, proficiency and potency of *Nigella sativa* in diverse medicinal sectors.

Key words: *Nigella sativa*, green synthesis, thymoquinone, Kalonji seed.

12.1. Introduction

Herbal medicines are used for curing diseases as they are safe when compared to modern allopathic medicines from many centuries in different indigenous and folk systems of medicines. Many researchers are focusing on medicinal plants since only a few plant species have been thoroughly investigated for their medicinal properties, potential, mechanism of action, safety evaluation and toxicological studies. *Nigella sativa* commonly called black seed cultivated mainly in Middle Eastern Mediterranean region, South Europe, India, Pakistan, Syria, Turkey, Saudi Arabia [1] and the seed oil is widely used as significant drug in the traditional medicine like Unani and Ayurveda [2-3]. *Nigella sativa* is having wide range of biological activities and therapeutic potential and it possess wide spectrum of activities such as diuretic, antihypertensive, antidiabetic, anticancer and immunomodulatory, analgesic, antimicrobial, antihelminthic, analgesics and anti-inflammatory, spasmolytic, bronchodilator, gastroprotective, hepatoprotective, renal protective and antioxidant properties. The seeds of *Nigella sativa* are widely used in the treatment of various diseases like bronchitis, asthma, diarrhea, rheumatism and skin disorders. It is also used as liver tonic, digestive, anti-diarrheal, appetite stimulant, emmenagogue, to increase milk production in nursing mothers to fight parasitic infections, and to support immune system. Most of the therapeutic properties of this plant are due to the presence of thymoquinone which is a major active chemical component of the essential oil [4,3].

12.1.1. Description of plant

Nigella sativa commonly called as black seed is an annual flowering plant (**Figure 12.1A**) belongs to the family Ranunculaceae. Flowers are solitary pale blue and white, with five to ten petals. The fruit is a large and inflated capsule composed of three to seven united follicles, each containing numerous seeds which are used as spice. Macroscopically, seeds are small dicotyledonous, trigonus, angular, regulose-tubercular, black externally and white inside with bitter taste and odour slightly aromatic. *Nigella sativa* and its seeds (**Figure 12.1B**) popularly known as ‘black caraway’, ‘black cumin’, ‘fennel flower’, ‘nigella’, ‘nutmeg flower’, ‘Roman coriander’, and ‘kalonji’.

12.1.1.1. Chemical Composition of *Nigella sativa* seeds:

Nigella sativa seed has many active important compounds that have been isolated, identified and reported so far like thymoquinone (30 % - 48 %), thymo-hydroquinone, di-thymoquinone, p-cymene (7 % - 15 %), carvacrol (6 % - 12 %), 4-terpineol (2 % - 7 %), t-anethol (1 % - 4 %),



Figure 12.1: Pictorial representation of *Nigella sativa* A) Flower B) Seeds

(Source: Adopted from internet)

sesquiterpene longifolene (1 % - 8 %) α -pinene and thymol *etc.* Seeds contain two different types of alkaloids; *i.e.*, isoquinoline alkaloids *e.g.*, nigellicimine and nigellicimine-N-oxide, and pyrazol alkaloids or indazole ring bearing alkaloids which include nigellidine and nigellicine [Figure 12.2] [5-6].

Seeds also contain other compounds *e.g.*, carvone, limonene, citronellol in trace amounts. The seeds of *Nigella sativa* contain protein (26.7 %), fat (28.5 %), carbohydrates (24.9 %), crude fibre (8.4 %) and total ash (4.8 %). The seeds are also containing good amount of various vitamin and minerals like Cu, P, Zn and Fe *etc.* The seeds contain carotene which is converted by the liver to vitamin A. Root and shoot are reported to contain vinic acid [5,7]. *Nigella sativa* seeds consist of quinine constituents most abundantly, thymoquinone attributes to the most of the therapeutic applications. The seeds reported to contain a fatty oil rich in unsaturated fatty acids, mainly linoleic acid (50-60%), oleic acid (20%), eicodadienoic acid (3%) and dihomolinoleic acid (10%). Saturated fatty acids (palmitic, stearic acid) amount to about 30% or less. α -sitosterol is a major sterol, which accounts for 44% and 54% of the total sterols in Tunisian and Iranian varieties of black seed oils respectively, followed by stigma sterol (6.57-20.92% of total sterols) [8-9].

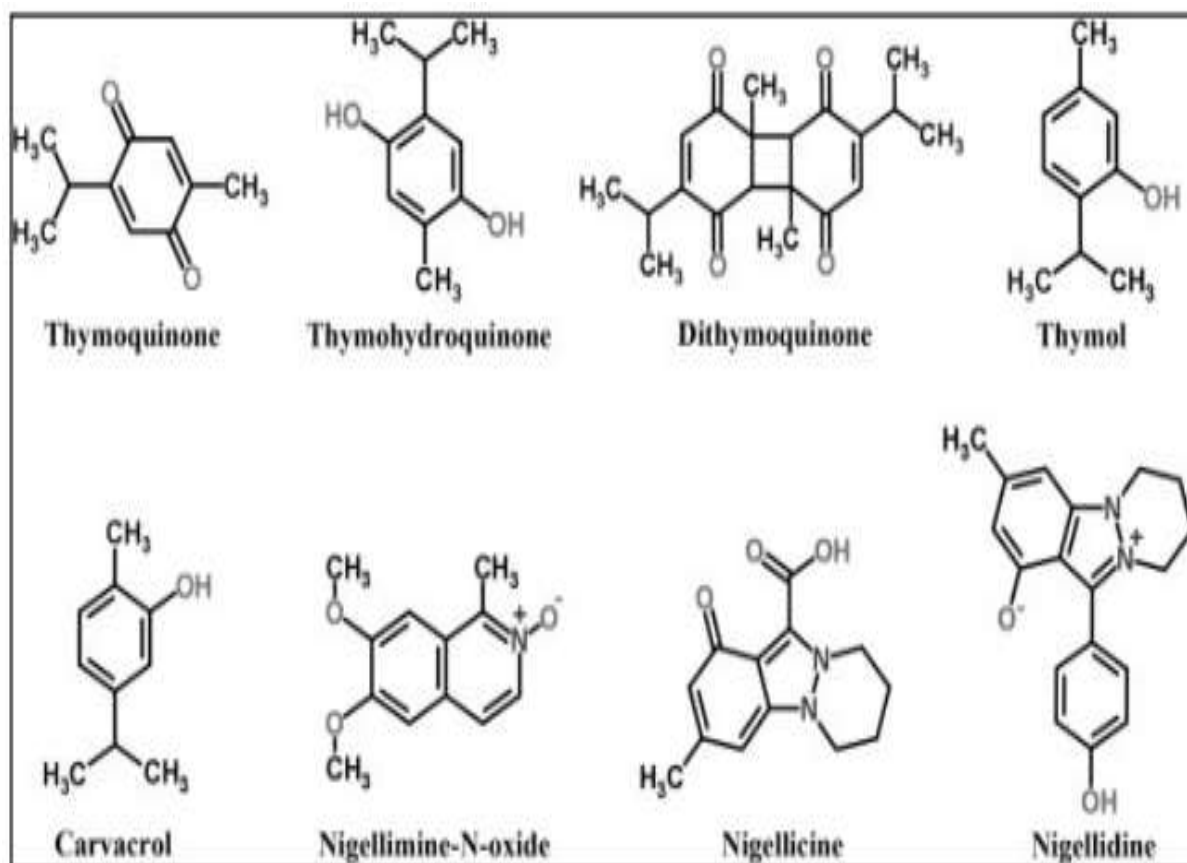


Figure 12.2: Pictorial representation of key chemical constituents of *Nigella sativa*

(Source: Adopted from internet)

12.1.1.2. Traditional and Medicinal Uses

The seeds are acrid, bitter, aromatic, thermogenic, carminative, diuretic, emmenagogue, anodyne, antibacterial, anti-inflammatory, digestive, appetizer, antithelminthic, febrifuge, stimulant, galactagogue, expectorant and sudorific. The seeds are also used for cough, amenorrhea, flatulence, jaundice, dyspepsia, inflammation, paralysis, skin, eyes, respiratory, stomach and liver problems, diarrhoea, dysentery and dysmenorrhoeal.

12.2. Therapeutic Applications of *Nigella Sativa* Seeds

The extensive research has been done on black seeds using modern scientific techniques. This plant is believed to be a miraculous herb that has potential to cure multiple ailments and disorders and their probable potential therapeutic applications are displayed in **Figure 12.3** and few of them are discussed below.

12.2.1 Anti-microbial activity

Crude extracts of *N. sativa* exhibited good effect against some of the tested organism. Gram

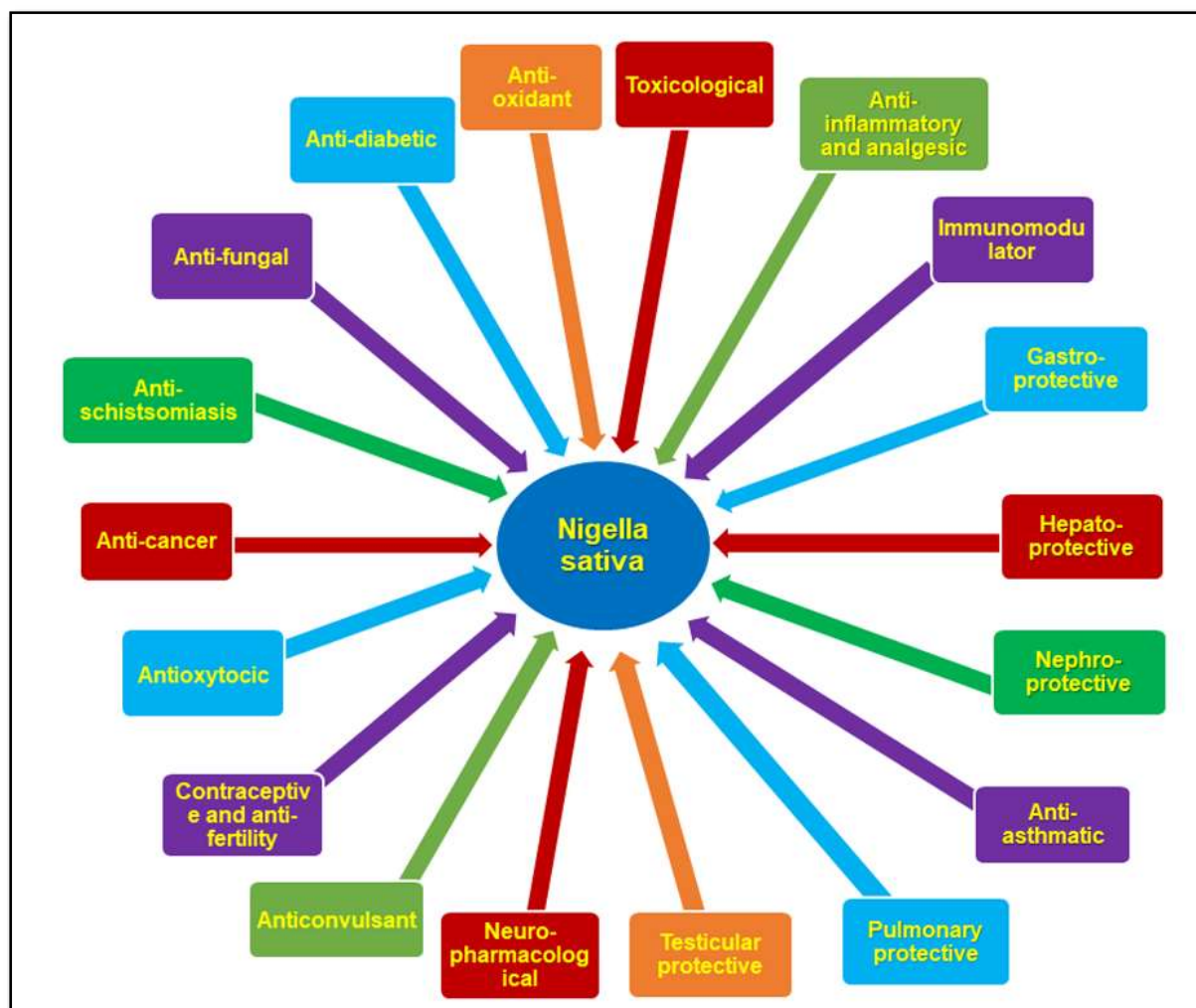


Figure 12.3: Therapeutic applications of *Nigella sativa*

negative bacteria was affected more than the gram positive [10]. The antibacterial activity of black seeds extract on *S. aureus* was monitored and confirmed by using the positive control Azithromycin [11]. Antibacterial activity of *N. sativa* against and triple therapy in eradication of *Helicobacter pylori* in patients with non-ulcer dyspepsia was carried out. Clinically useful anti *Helicobacter pylori* activity was observed with *N. sativa* seeds comparable to triple therapy [12].

Promising antifungal effect was evaluated with methanol followed by chloroform extracts of black seeds in different *Candida albicans* strains [13]. Potential anti-dermatophyte activity of ether extract of *N. sativa* and thymoquinone was tested against 8 species: four species of *Trichophyton rubrum* and one each of *T. interdigitale*, *T. mentagrophytes*, *Epidermophyton floccosum*, and *Microsporum*. The results denote the potentiality and supports its use in folk

medicine for the treatment of fungal skin infections [14]. Kalonji seeds quinines, di-thymoquinone, thymohydroquinone, and thymoquinone was examined *in-vitro*. Despite thymoquinone and thymoquinone of Kalonji seeds exhibited significant anti-yeast activity [15].

12.2.2 Anti-schistosomiasis activity

Mahmoud et al [16] studied the effect of *Nigella* seed oil on liver damage caused by *Schistosoma mansoni* (S. mansoni) in mice. The result suggest that *Nigella* seed oil may play a role against the alterations caused by *S. mansoni*. The antioxidant and anti-schistosomal activities of the garlic extract (AGE) and *Nigella* seed oil (NSO) on normal and *Schistosoma mansoni*-infected mice was investigated. Result showed that, protection with AGE and NSO prevented most of the hematological and biochemical changes and markedly improved the antioxidant capacity of *schistosomiasis* mice compared to the infected-untreated ones. These results suggested that garlic extract and *Nigella* seed oil are the potential protecting agents for the treatment of *schistosomiasis* infection [17].

12.2.3 Anti-oxidant activity

The antioxidant, anti-inflammatory, anti-cancer and anti-bacterial activities of the shoots, roots and seeds methanol extracts from *N. sativa* were studied. The three organs exhibited strong antioxidant activity using the oxygen radical absorbance capacity method and a cell-based assay. Bourgou et al [9] suggest the suppression of oxidative stress was observed in Wistar rats by thymoquinone. It also suggests that dietary supplementation of black seeds powder inhibits the oxidative stress caused by oxidized corn oil in rats [18]. It was also reported that oral feeding on the diet containing black seed powder at 10 % level antagonized the oxidative stress effects induced by hepato-carcinogens like di-butylamine and sodium nitrate (NaNO₃) in Swiss albino rats by normalizing reduced glutathione (GSH) and NO levels [19]. The black seed oil and thymoquinone by intra-peritoneal injection were found to shown protective effects on lipid peroxidation process during ischemia-reperfusion injury (IRI) in rat hippocampus [20].

Treating broiler chicks with black seed for six weeks prevented the liver from oxidative stress by increasing the activities of enzymes such as myeloperoxidase, glutathione-S-transferase, catalase, adenosine deaminase and by decreasing hepatic lipid peroxidation [21]. Significant *in-vitro* antioxidant activity of methanolic extract of *Nigella* was monitored by Mariod et al [22].

12.2.4. Anti-diabetic activity

Antidiabetic activity was monitored in diabetes induced rats with Kalonji seed extract and oil by Salama et al [23]. Treatment of rats with *N. sativa* extract and oil as well as thymoquinone significantly decreased the diabetes-induced increase in tissue MDA and serum glucose and significantly increased serum insulin and tissue SOD. The *N. sativa* oil restored normal insulin levels, but failed to decrease serum glucose concentrations to normal. In 2010, Abdel merguid et al [24] research on biochemical and ultra-structural findings suggests that *N. sativa* and thymoquinone are proved to be clinically useful in the diabetes treatment by pancreatic β cell integrity thereby decreasing oxidative stress.

The anti-hyperglycemic potential of thymoquinone on the activities of key enzymes of carbohydrate metabolism in streptozotocin (STZ)-nicotinamide (NA)- induced diabetes rats was evaluated. Oral administration of thymoquinone at 20, 40, 80 mg/Kg body weight for 45 days, dose dependently improved the glycemic status in STZ-NA induced diabetic rats. The levels of insulin, Hb increased with significant decrease in glucose and HbA (1C) levels [25]. The *N. sativa* showed the synergistic effect with human parathyroid hormone in improving bone mass, connectivity, biomechanical behavior and strength in insulin-dependent diabetic rats and found to be more effective as compared to the treatment with *N. sativa* or human parathyroid hormone alone [26]. *N. sativa* oil is effective in insulin resistance syndrome of diabetic and dyslipidemic patients [27].

12.2.5. Anti-inflammatory and analgesic activity

By applying the three different nociceptive experiments on the rats and mice i.e., hot plate method, tail-pinched method and acetic acid-induced writhing test it was concluded that the plant has ability to reduce the pain stimulus. The findings have confirmed that the plant black seeds fixed oil has potential anti-nociceptive actions that are may be due to an opioid part in the oil. This portion act as naloxoneis i.e., an inhibitory effect for the pain. Moreover, the plant oil has considerable CNS depressing activities [28].

12.2.6. Gastro-protective activity

Nigella seed oil and thymoquinone are proved to be protective against gastric lesions and alcohol induced lipid peroxidation [29]. Thymoquinone protects gastric mucosa against ulcers induced alcohol.

12.2.7. Hepato-protective activity

Administration of *Nigella sativa* protects liver tissue from toxic metals like lead and removes lipid peroxidation generated due to exposure to carbon tetrachloride [30]. Protection of rat liver against hepatic ischemia was determined by measuring biochemical parameters [31]. Fluconazole induced toxicity in wistar rats as reduced by the administration of *Nigella sativa* seeds. Black seeds restored the liver and kidney injury caused by fluconazole [32].

12.2.8. Nephro-protective activity

Nigella seed oil and thymoquinone reversed methotrexate-induced nephrotoxicity in albino rats by decreasing creatinine, preventing degenerative changes in kidney tissues [33-34]. Administration of NSO to Wistar rats exhibited protection against urolithiasis by decreasing urinary calcium, phosphate and acetate [35]. Hepato-nephroprotective activity of *Nigella* seed oil was demonstrated on paracetamol-induced rabbits. NSO maintained the shape of terminal hepatic venule (THV)/Central vein (CV) and sinusoids in the liver and normal shape of glomerulus and bowmans capsule in the kidney of Newzealand rabbit (*Oryctolagus cuniculus*) was observed [36].

12.2.9. Pulmonary protective activity and anti-asthmatic effects

The beneficial effects of NSO on rats with hyperoxia-induced lung injury were evaluated since oxygen-induced lung injury is believed to lead to the development of broncho-pulmonary dysplasia in premature infants. NSO significantly reduced the severity of lung damage due to hyperoxia [37]. The protective effect of *Nigella sativa* on tracheal responsiveness (TR) and lung inflammation of sulfur mustard GAS exposed guinea pigs was examined [38].

12.2.10. Testicular protective activity

Ethanollic extracts of Klonji seeds reduced the reproductive toxicity induced by fluoride. Aluminum in albino male mice. Fluoride and aluminum decreased the body weight, variation biochemical parameters like SOD, CAT, GSH levels in testis and epididymis, administration of kalonji seeds ameliorated toxicity [39]. Oxidative damage and tissue injury induced by formaldehyde in testicles may be prevented by using NSO which is a potential antioxidant [40]. Protective role of antioxidant property was reported against acetamiprid induced reproductive toxicity in male rats [41].

12.2.11. Neuro-pharmacological activity

Defatted infusions (aqueous and methanol) of *Nigella Sativa* seeds shows a commanding anti-depressant and analgesic activity. Elongated management of *Nigella Sativa* boost learning and memory whereas its oil and thymoquinone are used in the treating anxiety [42-43]. In addition to these, NSO has its strong effect in tramadol tolerance and dependence. *Nigella Sativa* extracts such as Aqueous and hydroalcoholic have neuroprotective effects in cerebral ischemia and its supplements progresses Pb-induced neurotoxicity, antioxidant and thymoquinone helps against chemotherapy-induced neurotoxicity [44-46].

12.2.12. Anti-convulsant activity

Both *N. sativa* oil and Curcumin has potential effect on anti- convulsant and antioxidative property by dropping induction of seizures, excitability and oxidative stress in epileptic animals as well as removers from some of the contrary effects of antiepileptic drugs. NSO, seeds, aqueous infusion and Thymoquinone has antiepileptic effect when pass through maximal electroshock (MES)- induced convulsions and minimal neurological deficit (MND) test [44].

12.2.13. Contraceptive and anti-fertility activity

The ethanolic extract of *Nigella sativa* seeds was found to possess an anti-fertility activity in male rats which might be due to inherent estrogenic potential. Oral administration of hexane infusion, their seeds prevent pregnancy in Sprague-Dawley rats as per reports of [44,47].

12.2.14. Anti-oxytotic activity

Nigella sativa seeds and its oil helps in inhibition of smooth muscle contraction of uterus induced by oxytocin stimulations and has anti- oxytotic effect in rats and guinea pig. [44].

12.2.15. Toxicological studies

Many toxicological studies related to *Nigella Sativa* seeds shows no toxicity whereas its oil has acute disorder. Moreover, the histo-pathological results also showed to be normal for the tissues of heart, liver, kidney, and pancreas [48]. Some studies pertinent to thymoquinone demonstrates the safety, particularly when given orally to experimental animals [44, 49].

12.2.16. Drugs-nigella interaction

Several studies demonstrated the potential of *Nigella sativa* interaction with co-administrated drugs and affect their intestinal availability and pharmacological effect. *In-vitro* studies reveals

CHAPTER 12: Therapeutic Applications of *Nigella Sativa* and its Metal Nanoparticles

that the extracts of *Nigella sativa* inhibit cDNA-expressed human cytochrome P-450, 3 A4, 2 C9, 3 A5 and 3 A7-mediated metabolism of marker substrates they may influence the affect and /or inhibit the metabolism of a wide range of drugs [50].

12.3. Green Synthesis of *Nigella Sativa* Nanoparticles and their Applications

Nanoparticles are small particles ranging between 1-100 nm in size and exhibit new properties based on their size, distribution and morphology. Research and development in nanotechnology have been growing rapidly around the globe. Nanoparticles are synthesized from metals by both chemical and biological methods. Despite to chemical synthesis, an alternative, eco-friendly and advantageous approach for synthesizing nanoparticles is the biological strategy which are synthesized from microbes (*Aspergillus flavus*, *Phomaexi guapseudomonas* species), and plants such as *Chenopodium album*, *Acalypha indica*, *Diopyros kaki*, *Cynodon dacylon*, *Glycyrrhiza glabra*, *Nigella sativa* etc.

Silver nanoparticles (AgNPs) were synthesized from the kalonji seed extracts and they were characterized by using several techniques like UV Spectroscopy, FTIR, XRD, SEM-EDAX, TEM. The synthesized AgNPs were used for varied applications like antibacterial, anti-cancer, anti-inflammatory, and antidiabetic activity. Likewise, gold nanoparticles (AuNPs) were synthesized from the black seed extracts and their applications were studied such as antibacterial, anti-biofilm, and cytotoxicity and antioxidant activity. *Nigella sativa* oil was used for the synthesis of Fe₃O₄ NP and they were characterized by UV-Vis., FTIR, IRM and SEM. Iron oxide nanoparticles (Fe₃O₄ NPs) were used for the antimicrobial studies. MnFe₂O₄/BC NP of kalonji seeds were prepared by Siddiqui et al [51] and they characterized and evaluated the antibacterial activity. Zinc oxide nanoparticles (ZnO NPs) were synthesized and they were used as food preservative material which inhibits Quorum sensing and as photocatalytic, antibacterial and antioxidants. Copper oxide nanoparticles (CuO NPs) were synthesized from black cumin seeds and their antibacterial potential was evaluated. A list was prepared and summarized the green synthesis of *Nigella sativa* and their nanoparticles application in Table 12.1.

12.4. CONCLUSION

Herbal medicines can be used as drugs in traditional medicine and gain popularity globally. A lot of studies existed on *Nigella sativa* which validate its importance in folk medicine as antihypertensive, anti-inflammatory, immune stimulant, analgesic, anti-microbial, anti-fungal,

Table 12.1: Nano metals of *Nigella sativa* and their applications

Sl.no	Plant part	Type of metal Np	Characterization techniques	Applications	Ref.
1.	Seed	Ag NPs	UV-visible, XRD, FTIR and TEM	Anti-inflammatory, antioxidant effects (Diabetic neuropathy)	[52]
2.	Seed	Ag NPs	UV-visible, XRD, FTIR and TEM	Antibacterial activity	[53]
3.	Seed	Ag nanocomposites		Hepatocellular carcinoma	[54]
4.	Seed	MnFe ₂ O ₄ /BC NP	FTIR, XRD, SEM, TEM, TGA and DSC	Antibacterial	[51]
5.	Nigella Sativa oil	Fe ₃ O ₄ NP	TEM, FTIR, IRM, SEM	Antimicrobial	[55]
6.	Seed	ZnO NP	XRD, SEM, Ultrasonic Irradiation	-	[56]
7.	Seed	ZnO NP	UV-visible, XRD, FTIR and TEM	Food preservative material inhibits Quorum sensing	[57]
8.	Seed	TiO ₂ @AG	TEM, SEM, DLS	Cytotoxicity	[58]
9.	Seed	ZnO NP	XRD, SEM, UV-visible	Photocatalytic, antibacterial, and antioxidant	[59]
10.	Seed	CuO NP	PXRD, SEM, UV-visible	Antibacterial	[60]
11.	Seed	AgNPs (nanorods)	FTIR, SEM, UV-visible	Antidiabetic	[61]
12.	Seed	Au NPs	FTIR, XRD, TEM, UV-visible	Antibacterial, antibiofilm, cytotoxicity	[62]
13.	Leaf	AgNPs	SEM, FTIR	Phytotoxicity, cytotoxicity, biofilm	[63]
14.	Seed	Au NPs	XRD, EDAX, TEM, UV-visible	Antioxidant activity	[64]

CHAPTER 12: Therapeutic Applications of *Nigella Sativa* and its Metal Nanoparticles

anti-oxidant, and as an anti-cancer. Further work can be desired to discover the pharmacokinetics, biochemical and therapeutic activities of active constituents and their collaboration with modern drugs. Cellular and molecular target of various constituent of *N. sativa* seed may be explored to get more effective and safer drug. Detailed studies are required to ensure its safety and efficacy and possible use of phytochemicals as drug candidate. Further thorough clinical studies on *N. sativa* should be done so that humanity can be benefited.

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Conflict of Interest

The authors declare that they have no conflicts.

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